

Tutorial 12

Ques1 Check whether the following are group? Justify your answer.

- a. $(\mathbb{Z}, +)$
- b. $(\mathbb{N}, +)$

Ques2 Show that set of all integers form a group w.r.t. binary operation defined by

$a * b = a + b + 1$, for all $a, b \in \mathbb{Z}$ is an abelian group.

Ques3 Show that the set of all positive rational numbers form as abelian group under the composition defined by $a * b = \frac{ab}{2}$.

Ques4 Let $S = \{0,1,2,3\}$ be a set with a binary operation defined by

$$a * b = (a + b) \bmod 4$$

Examine whether $(S,*)$ forms a semigroup and a monoid.

Ques5 Show that the identity e in a group always unique.

Ques6 Show that the inverse of each element of a group is unique.

Ques7 Determine whether the group $(\mathbb{Z}_6, +)$ is cyclic. Also, check whether, $(\mathbb{Z}_6, +, \times)$ is a ring.

Ques8 Prove that every cyclic group is an abelian group.